Hard vs. Soft Constraint PINN Project

- What I have worked on
- Current progress
- Plan for finishing the project from the blueprint

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Project question

The comparison is about accuracy, physical validity, uncertainty, and compute cost.

Main question

Does guaranteeing thermodynamic validity by construction outperform penalizing violations, and what does it cost in accuracy, calibration, and computational effort?

Evaluation focus	**What we check**
Accuracy	Prediction error on measured methane adsorption
Physical validity	Bounds, monotonicity, Sips shape, and Arrhenius checks
Extrapolation	Performance on withheld coal rank and high pressure
Uncertainty	Whether prediction intervals match observed errors

Data pipeline

The first milestone was a leakage-safe split with fixed evaluation tiers.

Split	Samples	Experiments
Full dataset	993	114
Training pool excluding T2	752	86
Training	472	54
Validation	132	14
T1 interpolation	148	18
T2 coal-rank extrapolation	241	28
T3 pressure extrapolation	3	2

Key pipeline choices

- Experiment-level grouping
- Train-only scaling
- Raw positive pressure preserved
- T2 holds out Subbituminous / Lignite

Note: T3 is counted inside T1, so it is not added as an independent split.

Blueprint progress

The implementation is mostly complete. The remaining work is stronger runs and final writing.

Blueprint part	Status	Output
Steps 1–2	Completed	Data pipeline and unified evaluation harness
Steps 3–8	Implemented	Soft baseline and hard model variants
Steps 9–11	Implemented	Ablation, UQ calibration, interpretability
Current work	In progress	Results analysis and report refinement
Next work	Planned	Final checking, writing, and packaging

Completed training protocol

The full V1 tuning run keeps the intended model size and uses the full cross-validation protocol.

Full protocol now completed

```
hidden_dims = (512, 1024, 512, 256, 128)
max_epochs = 1000
patience = 100
lambda_p_values = (0.0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0)
seeds = (0, 1, 2)
max_folds = 5
```

What this means

- 15 validation runs per λ_p
- 120 V1 tuning runs in total
- Final V1 checkpoint uses $\lambda_p = 0.05$

The remaining work is interpretation, report polishing, and reproducibility checking.

Model variants

The models compare soft penalties with hard architectural constraints.

Variant	Constraint design	Purpose
V1	All soft penalties	Reference model
V3	Hard bounds	Force adsorption below learned capacity
V4	Hard pressure monotonicity	Force adsorption to increase with pressure
V5	Hard Sips gauge	Build the pressure–adsorption curve into the model
V6	Hard Van't Hoff / Arrhenius	Constrain temperature dependence
V7	Hard bounds + monotonicity	Test interaction between two constraints
V2/V8	All-hard combined variants	Test maximum enforcement

V4 pressure isolation follows the blueprint: `pressure_MPa`, `Pr`, and `PT` are excluded from the free branch.

Soft-baseline selection

The soft baseline chooses λ_p using validation accuracy and physical validity.

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_p \mathcal{L}_{physics}$$
$$s_{val} = \mathrm{RMSE}_{val}\,(1 + v_{val})$$
$$v_{val} = \max\{\texttt{neg_rate}, \texttt{sat_rate}, \texttt{nonmono_rate}\}$$

s_{val} is the selection score. v_{val} is the largest validation rule violation.

λ_p	Val. score	Val. RMSE	Violation rate
0.00	10.349	5.174	1.000
0.05	**4.916**	**4.897**	0.004
0.50	4.969	4.966	**0.001**
1.00	4.941	4.922	0.004

The full run selects $\lambda_p = 0.05$. It keeps the validation RMSE low while reducing the validation violation rate sharply.

Full protocol RMSE results

Lower values mean smaller adsorption prediction error.

Variant	T1 RMSE	T2 RMSE	T3 RMSE
V1 soft baseline	5.300	5.746	4.943
V3 hard bounds	5.624	5.903	2.217
V4 hard monotonicity	**5.213**	5.709	4.643
V5 hard Sips gauge	5.410	5.476	1.372
V6 hard Van't Hoff / Arrhenius	6.082	5.543	5.478
V7 hard bounds + monotonicity	7.116	6.145	**1.192**
V2/V8 all-hard combined	6.209	**5.056**	1.435

Current pattern: V4 is best on T1, V2/V8 is best on T2, and V7 is best on T3. The V1 row uses the selected $\lambda_p = 0.05$ model. T3 has only 3 samples.

Physical validity and trade-offs

Hard constraints remove selected violations by construction.

Check	Current result	Interpretation
V4 monotonicity	Minimum slope 0.111; violation rate 0	Pressure isolation matches the blueprint
V5 Sips gauge	Predictions stayed physically valid	Pressure–adsorption shape is built in
V6 Arrhenius	Identity error 1.82×10^{-12}	Temperature relation holds numerically
V2/V8 all hard	Sips checks pass	All checked Sips rules pass

Trade-off to discuss

V4 satisfies the monotonicity check and performs best on T1, but no hard variant wins every tier. This supports a careful conclusion rather than a broad claim that hard constraints always improve accuracy.

Uncertainty calibration

Lower ECE means the predicted uncertainty matches the observed errors better.

Variant	Val. ECE	T1 ECE	T2 ECE	T3 ECE
V1 soft baseline	0.032	0.027	0.273	0.120
V3 hard bounds	0.042	0.053	0.242	0.120
V4 hard monotonicity	0.039	0.026	0.321	0.120
V5 hard Sips gauge	0.041	0.032	0.276	0.120
V6 hard Van't Hoff / Arrhenius	0.045	0.036	0.308	0.120
V7 hard bounds + monotonicity	**0.017**	0.015	0.299	0.120
V2/V8 all-hard combined	0.026	**0.015**	**0.202**	0.120

- Uncertainty estimates are less reliable on T2. V2/V8 has the lowest T2 ECE in the full run.
- T3 ECE is included for completeness, but it has only 3 samples.

Conclusion based on the full protocol

Current answer

Hard constraints help most clearly with physical validity. Their accuracy and uncertainty benefits depend on the model variant and evaluation tier.

- Physical validity: hard variants remove the targeted violations by design.
- Accuracy: V4 is best on T1, V2/V8 is best on T2, and V7 is best on T3.
- Calibration: V2/V8 has the lowest T2 ECE, while V7 has the lowest validation ECE.
- Next step: finalize the report discussion and reproducibility package.

Plan for the rest of the project

The main remaining work is writing, checking, and packaging.

Work	Why it matters	Expected progress
Verify outputs	Confirms tables match saved CSV files	Mostly complete after the full run
Finish report draft	Summarizes methods, results, and limits	Results discussion needs final polishing
Supervisor review	Checks scientific interpretation	Useful before final manuscript writing
Final package	Makes the project reproducible	Clean results, README, figures, and report